perplexity

Conditional Distribution I

Core intuition: a conditional distribution answers, "Once X is known, how should the uncertainty about Y change?" Think of starting with a joint distribution over (X, Y), taking the slice where $X = x$, then rescaling that slice so its total probability is 1.[1]

What to know first

You will usually begin with a **joint distribution**, which describes two variables together.

- **Joint:** $p_{X,Y}(x, y)$ or $f_{X,Y}(x, y)$: how likely the pair (X, Y) is.
- **Marginal:** $p_X(x)$ or $f_X(x)$: the distribution of X alone, after summing or integrating out Y.
- **Conditional:** $p_{Y|X}(y \mid x)$ or $f_{Y|X}(y \mid x)$: the distribution of Y when $X = x$ is treated as known.[2] [3]

For discrete variables, marginalize by summing:

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

For continuous variables, marginalize by integrating:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy$$

Part 1: likely foundations

Expect this first portion to focus on constructing and interpreting conditional pmfs/pdf's.

Discrete case

When $p_X(x) > 0$, use the **conditional-pmf formula**:

[
p_{Y\mid X}(y\mid x)

\frac{p_{X,Y}(x,y)}{p_X(x)}
]

This is the usual conditional-probability rule applied to random variables. The denominator is the total probability in the selected $X = x$ slice, so dividing by it makes the conditional probabilities add to 1.[4] [2]

A reliable workflow:

1. Identify the support: which (x, y) pairs have nonzero joint probability?
2. Find the marginal $p_X(x)$ by summing over y.
3. Divide the joint probability by that marginal.
4. Check that, for each fixed x, $\sum_y p_{Y|X}(y \mid x) = 1$.

Continuous case

For a joint pdf, the conditional density is

$$
[
f_{Y\mid X}(y\mid x)
\frac{f_{X,Y}(x,y)}{f_X(x)}
]
$$

provided $f_X(x) > 0$.[3] [2]

The notation "given $X = x$" can feel strange because a continuous random variable has probability zero of equaling one exact number. Do not interpret the formula as ordinary division by $P(X = x)$; it is a density-based conditional distribution, obtained by taking a vertical or horizontal slice of the joint density and normalizing it.[1]

The reconstruction identity

A central identity is:

$$
[
p_{X,Y}(x,y)
p_{Y\mid X}(y\mid x)p_X(x)
]
$$

or, for densities,

$$
[
f_{X,Y}(x,y)
f_{Y\mid X}(y\mid x)f_X(x)
]
$$

This matters because many statistical models are built in exactly this order: model X, then model the distribution of Y conditional on X.[5]

Part 2: likely extensions

The second section commonly builds from “find the conditional distribution” to “use it to compute quantities and reason about dependence.”

- **Conditional CDF:** $F_{Y|X}(y \mid x) = P(Y \le y \mid X = x)$.
- **Conditional expectation:** $E[Y \mid X = x]$, the conditional mean of Y at a fixed value of X.
- **Conditional variance:** $\mathrm{Var}(Y \mid X = x)$, the remaining spread in Y after X is known.
- **Independence:** if X and Y are independent, conditioning does not change the distribution:

$$
[
f_{Y\mid X}(y\mid x)
$$

f_Y(y)
]

Equivalently, the joint density or pmf factorizes into marginals.[2]

- **Bayes’ rule:** reverse the conditioning direction:

$$
[
p_{X\mid Y}(x\mid y)
$$

\frac{p_{Y\mid X}(y\mid x)p_X(x)}{p_Y(y)}
]

Bayes’ rule becomes especially important later in statistical inference: prior distribution $\times$ likelihood, normalized into a posterior distribution.[6] [4]

A useful mental model

For your ML background, frame this as a **generative factorization**:

$$p(x, y) = p(y \mid x)p(x)$$

For supervised learning, $p(y \mid x)$ is often the object of greatest interest: it describes the target’s uncertainty after observing features. Regression estimates a conditional mean such as $E[Y \mid X = x]$; probabilistic classification estimates conditional class probabilities such as $P(Y = k \mid X = x)$.

Before class, try explaining in your own words why the denominator in $p_{Y|X}(y \mid x)$ must be $p_X(x)$, rather than $p_Y(y)$.

⁂

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3. https://en.wikipedia.org/wiki/Marginal_distribution

4. https://arxiv.org/html/2603.27038v1
5. https://stats.stackexchange.com/questions/485156/what-are-the-differences-between-marginal-probability-distribution-and-condit
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